Akoizo Original Practice - Form 4 - Answer key

Division C | Experimental Design | 50-minute written practice

Original Akoizo curriculum practice. This is not a historical tournament paper. Written build, CAD, lab and communication exercises supplement real practical preparation.

ANSWER KEY - Review only after submitting.

1. (2 points) Sample SD for 8,10,12 is…

A. 10

B. 8

C. 2

D. 4

Correct: C

2. (2 points) A pilot primarily checks…

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

A. Whether to delete awkward results

B. The final conclusion automatically

C. Every possible mechanism

D. Feasibility and procedure

Correct: D

3. (2 points) Which addresses time-order drift?

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

A. Deleting late trials silently

B. Randomization or suitable blocking

C. Only more decimals

D. Only a larger title

Correct: B

4. (2 points) A reproducible timing method needs…

Unit: Materials, reproducible procedures, and qualitative data.
Worked-case data: A trial has no timer reading because the observer missed the endpoint. What should the table and procedure record?

A. Only a final mean

B. Only a title

C. Explicit start and stop events

D. Only a stopwatch brand

Correct: C

5. (2 points) A useful practice reflection leads to…

Unit: Practice, timing, teamwork, and final review.
Worked-case data: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

A. A specific correction tested on a new task

B. Only repeating the same answer

C. Blaming all variation on humans

D. Removing the error section

Correct: A

6. (4 points) Write the example's claim and evidence.

Unit: Claims, evidence, and reasoning for trends and outliers.
Worked-case data: At levels 1,2,3, means are 12,16,20 units, with SD 2 at each level. Write a bounded trend CER.

answer: Mean response increases over levels 1-3; means are 12,16,20 with SD 2, an overall rise of 8 units. (4 points).

7. (4 points) Distinguish control condition and controlled variable.

Unit: Problem, hypothesis, variables, and experimentation.
Worked-case data: All low-level trials occur early and all high-level trials occur late while the instrument drifts upward.

answer: A control condition is a reference treatment; a controlled variable is held comparable across treatments. (4 points).

8. (4 points) Why does a control mean have uncertainty?

Unit: Division C comparison standards, significant figures, and abstracts.
Worked-case data: Report area from measured lengths 2.0 cm and 3.00 cm, then report their sum.

answer: It comes from measurements and sampled units with variability, not an exact definition. (4 points).

9. (4 points) Why show individual trials?

Unit: Quantitative data, graphs, and statistics.
Worked-case data: For response values 8, 10, 12, calculate mean, range, and sample SD.

answer: They reveal spread, unusual values, sample size, and patterns that a mean alone can hide. (4 points).

10. (4 points) What should be checked in the actual packet?

Unit: The experiment and the prompt packet.
Worked-case data: A prompt supplies a ramp, object, ruler, and timer. Turn “study motion” into a testable question.

answer: Topic and material limits, timing, safety, required sections, and the applicable scoring expectations. (4 points).

11. (4 points) Write a useful follow-up question.

Unit: Errors, conclusions, and recommendations.
Worked-case data: Every timer reading is 0.5 s too high. Would ten more repeats eliminate the bias?

answer: Test whether the observed level-response trend persists with randomized order and stable temperature, using the same operational response and independent replication. (4 points).

12. (2 points) Define “Resolution” in the context of Experimental Design.

answer: Smallest distinguishable instrument increment under the stated reading method. (2 points).

13. (6 points) Solve this case and explain your reasoning: A team finishes data collection with little time left and no graph or analysis. Suggest a targeted practice change.

reasoning: Measure where time went: setup, resetting, recording, or coordination.
Simplify the apparatus or parallelize independent tasks while preserving definitions.
Reserve analysis time and test the revised workflow on a new prompt. (3 points).

conclusion: The improvement addresses the actual bottleneck rather than sacrificing evidence quality. (3 points).

14. (2 points) Define “Claim” in the context of Experimental Design.

answer: Answer to the defined question. (2 points).

15. (2 points) Define “Controlled variable” in the context of Experimental Design.

answer: Condition kept comparable. (2 points).

16. (2 points) Define “Vehicle control” in the context of Experimental Design.

answer: Condition receiving a treatment carrier without the active component. (2 points).

17. (2 points) Define “Residual” in the context of Experimental Design.

answer: Observed value minus model prediction. (2 points).

18. (2 points) Define “Pilot” in the context of Experimental Design.

answer: Preliminary feasibility check before the final protocol. (2 points).